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Transforming social media sentiment analysis with generative large language models: A case study from the cultured meat debate

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ABSTRACT

Social media platforms can serve as important data sources for discourse about novel foods. However, prevailing methods for analyzing the food-related and other sentiments expressed on such platforms by their users have drawbacks including high cost, time intensiveness, and/or low accuracy. To address those drawbacks, this paper presents a series of sentiment-analysis frameworks leveraging the generative large language model (LLM) GPT-4o. Using a human-annotated subset of more than 230,000 cultured-meat tweets from a 12.5-year period, we compared five methods: i.e., three GPT-4o configurations (zero-shot with basic instructions, zero-shot with detailed guidelines, and fine-tuning), alongside VADER and BERT. Among them, fine-tuned GPT-4o demonstrated superior accuracy, while zero-shot GPT-4o, with or without detailed guidelines, achieved performance comparable to BERT without requiring any model training. Applying the LLM framework using fine-tuned GPT-4o to the full dataset as a case study captured meaningful dynamics within online discourse about cultured meat, including its overall balance of positivity, negativity, neutrality, and mixed views, and how regulatory milestones might have affected it. Overall, our rigorously evaluated, highly scalable generative LLM frameworks for sentiment analysis were able to accurately and efficiently process online discourse about a novel food – outperforming widely used lexicons (VADER) and close to or equaling the performance of fine-tuned deep-learning models (BERT) – without requiring extensive machine-learning expertise or even pre-labeled datasets in some instances. As such, they provide food and consumer scientists with easy-to-deploy and time-efficient computational tools for analyzing online narratives.

1. Introduction

Innovations in food technology are essential to achieving a more sustainable and robust food system (Giacalone & Jaeger, 2023). Yet, consumers are often hesitant to accept or consume novel foods (Siegrist & Hartmann, 2020). Given that public attitudes toward these innovations can influence the breadth of their adoption (Nucci & Hallman, 2015), it is vital for stakeholders to understand how the public perceives these technologies. However, prevailing methods of ascertaining public attitudes toward products can be costly, time intensive, and/or relatively inaccurate. This methodological paper addresses those drawbacks.

1.1. Measuring public opinion: Surveys and focus groups vs. social media data

Surveys have traditionally been the primary tool for collecting and analyzing public opinion (Reveilhac et al., 2022), closely followed by focus-group discussions (e.g., O'Keefe et al., 2016; Tucker, 2014). However, conducting surveys or focus groups involves substantial time and cost (Zhang et al., 2022), especially when they are cross-national and/or large in scale. Therefore, in recent years, increasing numbers of public-opinion studies have turned to data drawn from social media platforms, which offer convenient ways for individuals to share and discuss a wide range of topics (e.g., El Barachi et al., 2021; Hu et al., 2021). Also, because such platforms contain large quantities of global

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data generated over long periods (Garcia et al., 2021), using them as data sources can help address some of the time, space, and other limitations of traditional methods (Chen & Zhang, 2022). Among such platforms, Twitter/X, which had more than 368 million monthly active users as of December 2022 (Statista, 2023) and more accessible data than some other social media platforms, is favored for research in diverse fields, including food-related studies (e.g., Molenaar et al., 2024). While social media data do not represent the general population, they provide a window on public discourse that can complement more traditional methods of data collection (Reveilhac et al., 2022).

1.2. Computational sentiment-analysis approaches

Robust computational methods are required to derive insights from large volumes of text, particularly when it is being generated almost constantly on social media platforms. One such methodological domain is natural language processing (NLP), which aims to enable computers to understand, interpret, and potentially generate human language (Drikvandi & Lawal, 2023). Within NLP, sentiment analysis focuses on extracting emotions from text by classifying it: generally, as positive, negative, and neutral (Guo, 2022; Hardeniya & Borikar, 2016). Writing in 2022, Chen & Zhang (2022) noted that all published sentiment analyses of food-related social media data up to that time had used lexicon-based methods (e.g., Feldmeyer & Johnson, 2022; Mostafa, 2019; Trivedi & Singh, 2021). Such methods mainly rely on pre-defined dictionaries of words and associated sentiment scores (Hussain et al., 2023). One of the lexicon-based methods most commonly used in food-related studies is VADER (Valence Aware Dictionary for sEntiment Reasoning; e.g., Jenkins et al., 2023; Mohamed et al., 2023; Molenaar et al., 2024), which was specifically designed for analyzing social media text (Ainapure et al., 2023). Although lexicon-based methods are easy and convenient to deploy, as they do not require training data (Gomez-Adorno et al., 2024), they often have low accuracy due to limitations such as their lexicons' limited coverage and their inability to capture context-dependent meanings (Pratama & Hayaty, 2023; Sazzed & Jayarathna, 2021).

As noted by Joloudari et al. (2023) and Zhao et al. (2021), the 2018 introduction of an advanced deep-learning model, BERT (Bidirectional Encoder Representations from Transformers; Devlin et al., 2019), revolutionized NLP by achieving state-of-the-art performance across a range of tasks, including sentiment analysis. Because it has been pre-trained on a corpus of 3.3 billion words (Devlin et al., 2019), BERT has a rich understanding of how words relate to one another in various linguistic contexts (Sadler et al., 2024). To apply its general language understanding to specific tasks, however, BERT typically requires fine-tuning (i.e., additional training on a labeled dataset that is task-specific), which allows it to adapt its pre-trained knowledge to the requirements of a given task (Lohani et al., 2024). The identification of tweets' sentiments is one example of a task for which fine-tuning of BERT would be needed. However, obtaining the labeled data necessary for fine-tuning can be time-consuming and costly (Aouedi et al., 2022). For sentiment-analysis tasks, the labeling process typically involves human annotators – either domain experts or crowd-sourced workers – who manually identify and classify sentiments in text documents, often guided by a codebook (Van Atteveldt et al., 2021). Though its results may be consistent and reliable, human labeling of data poses scalability challenges: a severe drawback for sentiment-analysis models that must have large labeled datasets (Tan et al., 2023).

More recently, the development of generative LLMs such as the GPT series (e.g., ChatGPT; OpenAI, 2024b) has introduced a new approach to addressing diverse NLP tasks (Gray et al., 2023), including sentiment-analysis sub-tasks (Belal et al., 2023). GPT-4 has billions of parameters and is pre-trained on vast amounts of textual data, allowing it to generate human-like responses based on user-provided prompts (Tao et al., 2024; Tian et al., 2024). This ability enables generative LLMs to perform tasks through zero-shot (ZS) prompting, i.e., not requiring any task-specific labeled data and instead relying solely on instructions (Wang & Gayed, 2024). An example of a ZS prompt for a sentiment-analysis task is: "What's the attitude of the sentence X to the target t? Select one from: positive, negative, neutral," where X represents the text to be analyzed and t refers to the subject or entity in focus (Rusnachenko et al., 2024, p. 3152). This capability can vastly reduce the time and cost associated with acquiring and labeling the task-specific data that older machine-learning methods must have. However, ZS prompting is not the only available technique for LLM-based sentiment analysis. Generative LLMs such as GPT-3.5 and GPT-4 can also be fine-tuned for specific tasks, including sentiment analysis.

Some recent studies explored the potential of using generative LLMs for sentiment analysis, either through ZS prompting or fine-tuning, with some comparing both approaches (e.g., Belal et al., 2023; Fatouros et al., 2023). However, the prompts used in those studies lacked some key task-specific details of the kind normally found in coding guidelines for human-labeling work. For example, to analyze the sentiment of free-text responses to COVID-19 surveys, Lossio-Ventura et al. (2024) prompted ChatGPT with "What is the sentiment of the following sentence: 'x'?" But in the absence of precise, task-specific definitions of sentiment categories, generative LLMs will make sentiment judgments predominantly based on knowledge they acquired during their original developers' large-scale pre-training phases; and this is likely to reduce the interpretability and consistency of the results. Conversely, GPT-4 fine-tuned on task-specific data has been found to perform better at sentiment analysis than GPT-4 using ZS prompting with basic instructions (Roumeliotis et al., 2024). This implies that fine-tuning with labeled examples can enable such models to better internalize task-specific knowledge. Nevertheless, it remains unclear how well generative LLMs can interpret task-specific coding guidelines designed for human annotators, or whether incorporating such guidelines into their instructions would enhance their sentiment-analysis performance more than fine-tuning does. Also, to the best of our knowledge, no studies have leveraged the potential of generative LLMs for sentiment analysis of food-related social media data.

1.3. The present study

Accordingly, this study explored how GPT-4o, an enhanced version of GPT-4 and the latest available model in the GPT series when our main analyses began, could be applied to sentiment analysis of food-related online discourse. Our chosen target food for this methodological intervention is cultured meat, which is produced by culturing cells obtained non-lethally from an animal source (Gruber, 2022). Much cultured-meat opinion research has relied on traditional surveys and focus groups (Bryant et al., 2019; Bryant & Dillard, 2019; Wilks & Phillips, 2017). While this product is perceived by some consumers as a way to improve animal welfare and environmental sustainability (Rolland et al., 2020; Tuomisto, 2019), others express concerns about its safety, nutritional value, and perceived unnaturalness (Bryant & Barnett, 2020). Using a cultured meat-related Twitter/X dataset as an example, we 1) evaluated the performance of GPT-4o under three settings – ZS prompting with basic instructions, ZS prompting with a validated codebook, and fine-tuning with labeled data – and compared those settings' respective performance against those of VADER and BERT; and 2) applied the best-performing approach identified in step 1 to map longitudinal sentiment shifts across key cultured meat-related news events, demonstrating the method's capacity for monitoring online public discourse.

The contributions of this study are threefold. First, its detailed comparative evaluation of five sentiment-analysis methods offers critical insights for method selection in future research, as each has its own strengths and limitations. Second, it systematically demonstrates and compares three LLM-based approaches, each incorporating distinct prompting designs and training paradigms. By explicitly introducing codebook-based instructions grounded in human labeling practices, this approach advances existing ZS strategies that often overlook the

potential roles of human-guided coding frameworks in LLM-based analysis. Third, by validating the best-performing LLM-based framework on a large-scale longitudinal dataset, this study establishes an accessible and scalable methodological tool tailored to the food domain that can reliably capture changes in sentiment over time.

Collectively, these contributions provide both methodological advancement and practical guidance for researchers developing, testing, and/or applying LLM-based approaches to social media text analysis, particularly in contexts requiring rigorous, human-centered content analysis. In addition, these tools equip food scientists, policy communicators, and industry practitioners with flexible methodological options for monitoring the tone and dynamics of online narratives surrounding emerging food technologies, enabling more effective and evidence-based communication strategies during critical stages of food innovation.

2. Methods

2.1. Data collection and cleaning

Our initial dataset of 424,382 English-language tweets posted between July 1, 2010, and December 31, 2022 was collected via Brandwatch using the query: "cultured meat" OR "lab grown meat" OR "in vitro meat" OR "cell based meat" OR "cultivated meat" OR #culturedmeat OR #labgrownmeat OR #invitromeat OR #cellbasedmeat OR #cultivatedmeat. In selecting those search terms, we prioritized terminology that was both commonly used in academic, media, and industry discussions and that consistently yielded relevant tweets across the 2010–22 period. For further details of the data-collection process, please refer to T. Chen et al. (2024). We removed duplicate tweets (following Lavanya et al., 2020) and also excluded those that did not in fact contain any of the keywords in our query. This left 231,777 tweets for analysis.

2.2. Data labeling

To provide fine-tuning data for GPT-4o and BERT, as well as a human-coded baseline for evaluating the performance of this study's five sentiment-analysis methods, human-coded data were required. To ensure annotation consistency, we developed and validated a codebook, the finalized version of which is set forth in Table S1 of the Supplementary Materials. On Twitter/X, users may express their sentiments directly or simply share news or other information without necessarily conveying any sentiment. To differentiate sentiment-free tweets from those expressing both positive and negative sentiments, we defined the former as neutral tweets and included a "combination" category for the latter. Initially, we reviewed a randomly selected subset of tweets and developed a draft codebook (following Papautsky et al., 2021) that classifies them into four categories, namely positive, negative, neutral, and combination.

Before each round of coding discussion and refinement, three graduate students (including one of the authors) used the latest draft of the codebook to independently annotate a random subset of 200 tweets. We then checked inter-coder reliability, i.e., the consistency of sentiment labeling across coders in this case, using Krippendorff's α (Krippendorff, 2004). This metric was chosen because it is more versatile and accurate than either percent agreement or Cohen's κ (Cohen, 1960). Specifically, Krippendorff's α supports more than two annotators per sample and accommodates missing data, while still – like Cohen's κ – accounting for the possibility that agreement between annotators could occur by chance (Nili et al., 2020). After each round of labeling, the annotators discussed any disagreements, revised the codebook accordingly, and then used the revised codebook to annotate a new subset comprising 200 previously unlabeled tweets. This process was repeated until α's value reached 0.670. An α value above 0.667 is generally considered acceptable, and values above 0.8 indicate good agreement (Ishikawa et al., 2019). However, Salminen et al. (2018) found that annotation tasks in social computing, which are inherently subjective, typically achieve lower agreement scores than the usual threshold, often averaging around 0.6 for both κ and α. Based on those findings, we followed Smetanin (2022) in interpreting reliability scores ranging from 0.61 to 0.8 as representing substantial agreement, and scores from 0.81 to 1.0 as suggesting almost perfect or perfect agreement.

Following the establishment of substantial agreement, we constructed our final human-annotated baseline. As recommended by Tong et al. (2022), the sample tweets we used during codebook development and annotator training were excluded from the final labeling. This ensured that the annotators, who had already seen those tweets, would assign all further labels based on the finalized codebook rather than relying on their memories. From the remaining dataset, a subset of approximately 10% (n = 22,975) was then randomly sampled and assigned to the three annotators. Specifically, two of them labeled 7658 tweets each, and the third, 7659 tweets. Most of those 22,975 tweets were classified as neutral (64.7%; n = 14,876). Among the rest, positive tweets (23.9%; n = 5496) outnumbered both negative (9.3%; n = 2141) and combination ones (2.0%; n = 462).

2.3. Model comparisons

To facilitate our assessment of generative LLMs' effectiveness for sentiment analysis, we split the human-labeled dataset into a training set (80%, n = 18,380) and a testing set (20%, n = 4595). The training set was used to fine-tune BERT and GPT-4o, and the testing set was used to evaluate the performance of all five approaches.

2.3.1. Baseline models

VADER. VADER, one of the most popular lexicon-based sentiment-analysis methods (Mahrukh et al., 2023; Qi & Shabrina, 2023), extends beyond typical opinion words by including acronyms/initialisms (e.g., "LOL" and "ROFL"), slang (e.g., "Nah" and "meh"), and emoticons (e.g., ":)" and ":D") that are commonly found on social media platforms (Al-Shabi, 2020). VADER calculates a compound score for each document by summing the sentiment values of all words in it, then normalizing this score to a range between −1 (most negative) and +1 (most positive). Following the recommendations of VADER's developers (Hutto & Gilbert, 2014), we classified tweets with a compound score greater than or equal to 0.05 as positive, those with a score less than or equal to −0.05 as negative, and those with a score between −0.05 and 0.05 as neutral. Importantly, when this method is used, positive and negative words can cancel each other out, meaning that the overall sentiment of a tweet depends on the balance between these opposing sentiments. Therefore, a tweet identified by VADER as neutral may either lack strong sentiment indicators or contain a combination of positive and negative words that lead to a net neutral sentiment score. As such, tweets expressing both positive and negative sentiments – which our human annotators labeled as "combination" – cannot be distinguished by VADER.

BERT. Given BERT's strong performance across various NLP tasks, we selected it as our second baseline model for comparison. Specifically, we utilized the BERT-base-uncased model from Hugging Face (see https://huggingface.co/), a company that provides access to various pre-trained models (Miah et al., 2024). As noted by Rogers et al. (2020), a "pre-training + fine-tuning" workflow is not only a conventional paradigm but also a crucial foundation for BERT. Notably, fine-tuning BERT on task-specific data has been found to yield state-of-the-art results across diverse NLP tasks (Ma et al., 2019). Therefore, BERT was evaluated here in its standard high-performance configuration: fine-tuned on our task-specific training dataset mentioned in Section 2.3, above. Our hyperparameter settings for it, shown in Table 1, were chosen either because they had performed well in similar sentiment-analysis tasks (e.g., Jin et al., 2024; Lin & Moh, 2021) or because they were recommended by BERT's developers (Devlin et al., 2019). The remaining hyperparameters were kept at their default values.

Table 1
Hyperparameter settings for BERT.

Parameter	Value
Learning rate	2e-5
Batch size	16
Number of epochs	3
Optimizer	AdamW
Maximum sequence length	128

2.3.2. GPT-4o

Of the various GPT-series models, GPT-4o, released on May 13, 2024 (OpenAI, 2024a), was the newest when our key data explorations started in June 2024, and was therefore used as the baseline for the LLM frameworks in this study. As noted by Islam & Moushi (2024), GPT-4o can outperform its predecessors in processing speed and exhibits remarkable performance in text understanding and generation. Unlike BERT, GPT-series models such as GPT-4o can perform ZS sentiment classification directly from natural-language prompts (Roumeliotis et al., 2024) while also supporting fine-tuning. Accordingly, the next two subsections respectively explore the performance of GPT-4o with those two techniques.

ZS GPT-4o. By integrating instructions tailored for sentiment analysis (e.g., a codebook) and the target tweet into a prompt (i.e., ZS prompting), we can obtain an LLM's prediction of a tweet's sentiment despite not providing the model with any labeled data. We evaluated two ZS configurations: 1) ZS GPT-4o (no CB), referring to ZS GPT-4o without a codebook, i.e., a prompt with basic instructions, and 2) ZS GPT-4o (CB), referring to ZS GPT-4o with a codebook, i.e., a prompt containing the complete, finalized codebook designed for human annotators and established through iterative refinement and inter-coder reliability validation as described in Section 2.2 (Table 2).

The process of sending prompts to the model and obtaining its responses can be automated using OpenAI's application programming interface (API). Here, we used the OpenAI API to access the "gpt-4o-2024-08-06" model, which was the latest version of GPT-4o available as of October 2024 when we started actual data analysis and model evaluation (OpenAI, 2024a). For both the ZS GPT-4o configurations we tested, each prompt included a single tweet to be analyzed. Temperature – the parameter that controls the randomness of the model's response, with higher values resulting in greater randomness (OpenAI, 2024c) – was set at zero.

Fine-tuned GPT-4o. Fine-tuning GPT-4o involved providing it with our labeled data and training it to predict tweets' sentiments. We fine-tuned "gpt-4o-2024-08-06", the only version of GPT-4o available for that purpose as of October 2024, using the same training set we had previously used for BERT. More specifically, we converted the training set into a JSONL file containing prompt-completion pairs; and our prompt is shown in Table 2. Completion, which is not shown in Table 2, represents the human-labeled sentiment of a given tweet.

The training data for GPT-4o was then uploaded to OpenAI's cloud servers and used for fine-tuning. As recommended by OpenAI (2024c), we set the number of epochs, batch size, and learning rate multiplier as "auto", thus allowing the system to select their respective values based on the dataset's size. The final system-determined values for those three parameters are shown in Table 3.

Inference is the process of using a trained model to make predictions from new data (Sayadi & Aghbari, 2023). In the case of fine-tuned GPT-4o (hereafter, FT GPT-4o), inference refers to the process of applying it to new data that it was not exposed to during fine-tuning. When using FT GPT-4o for inference, we used the same prompt template as during fine-tuning (Table 2). The key difference is that in fine-tuning, the model was trained using labeled sentiment data, whereas in inference, it predicted sentiment labels for unencountered tweets. Temperature was again set to zero to minimize randomness.

2.3.3. Evaluation

In all five approaches, the task was to predict the sentiments of the tweets in the testing set, and a given approach's performance was evaluated by comparing the sentiment labels it generated against the human-annotated baseline. As mentioned previously, VADER cannot identify combination tweets. Therefore, when evaluating VADER's performance, we removed the 103 tweets labeled as combination by annotators from the testing data, leaving 4492.

BERT was set to evaluation mode, which ensures that the same input consistently produces the same output (Aboukass & Anoun, 2024). Similarly, VADER's rule-based design guarantees the same output for the same input. Therefore, these two approaches were run on the test data only once. In contrast, GPT models are non-deterministic, meaning that the same prompt can lead to different responses, even with the temperature set to zero (Ateia & Kruschwitz, 2023). To balance our need to account for this non-determinism against the large size of the testing dataset and corresponding API costs, we adopted a three-run evaluation approach, consistent with the methodology employed by recent studies (Li et al., 2024; Wu et al., 2024). Consequently, each of the three tested GPT-4o configurations was evaluated three times on the same test data using the same prompt. Then, for each of those three configurations, we calculated the average and standard deviation (SD) of their respective performance across the three runs. In addition, following Reiss (2023), we used Krippendorff's α to evaluate the intra-coder reliability of each GPT-4o configuration, i.e., whether the same input consistently produced the same outputs.

Following Ho et al. (2020), we used overall accuracy and weighted F1 score to evaluate the five approaches' performance. Overall accuracy was calculated as the ratio of correctly predicted instances to total instances: a straightforward measure of prediction success (Straube & Krell, 2014). Weighted F1 score, on the other hand, calculates F1 score for each category, then computes a weighted mean based on the

Table 2
Prompts used in all GPT-4o configurations.

Configuration	Role	Content
ZS GPT-4o (no CB)	System	You are an expert in conducting sentiment analysis.
	User	"Evaluate the sentiment toward cultured meat that this tweet expresses:" + {tweet} + "\n" "Return your response only as positive (which indicates positive sentiment toward cultured meat is expressed), negative (which indicates negative sentiment toward cultured meat is expressed), neutral (which indicates no sentiment toward cultured meat is expressed), or combination (which indicates that both positive and negative sentiments toward cultured meat are expressed)."
ZS GPT-4o (CB)	System	You are an expert in conducting sentiment analysis.
	User	"Here is a codebook that will guide your sentiment analysis of tweets about cultured meat:" + {codebook} + "\n" + "Perform sentiment analysis on the following tweet:" + {tweet} + "\n" + "Return your response only as one of the four coding numbers indicated in the codebook."
FT GPT-4o	System	You are an expert in conducting sentiment analysis.
	User	"Evaluate the sentiment toward cultured meat that this tweet expresses:" + {tweet}. + '\n' + "Return your response only as positive, negative, neutral, or combination."

Note. ZS = zero-shot prompted; FT = fine-tuned; {tweet} = the content of a tweet; {codebook} = the content of the codebook; \n = a line break.

Table 3
Hyperparameter settings for fine-tuning GPT-4o.

Parameter	Value
Learning-rate multiplier	2
Batch size	12
Number of epochs	1

number of samples in each category (Bozuyla, 2024). This metric was particularly useful in our case because our dataset can be considered imbalanced, due to more than 60% of the labeled tweets being classified as neutral.

Overall accuracy and weighted F1 score can be computed as

$$Overall\ accuracy = \frac{\sum_{i=1}^{n}(TP_i + TN_i)}{\sum_{i=1}^{n}(TP_i + TN_i + FP_i + FN_i)} \quad (1)$$

$$Weighted\ F1 = \sum_{i=1}^{N} \omega_i \left(\frac{2TP_i}{2TP_i + FN_i + FP_i} \right) \quad (2)$$

where N is the number of categories in the classification task (in this study, $N = 4$); i is the index representing each individual category in the summation, ranging from 1 to N; TP_i (true positive) is the number of samples for which the model correctly predicts that the sample belongs to category i; TN_i (true negative) is the number of samples for which the model correctly predicts that the sample does not belong to category i; FP_i (false positive) is the number of samples that the model incorrectly predicts as category i, when they actually do not belong to it; FN_i (false negative) is the number of samples that actually belong to category i but are incorrectly predicted as not belonging to it; and ω_i is the weight of category i, i.e., the number of samples in that category divided by the total number of samples. The weights for positive, negative, neutral, and combination tweets were approximately 23.6%, 9.2%, 64.9%, and 2.2%, respectively.

We also computed a normalized confusion matrix for each method. Such matrices, with rows representing actual labels and columns representing predicted labels, enable visual interpretation of results (Parkhi et al., 2012). Each row is normalized to sum to 1, and each cell in the matrix shows the proportion of samples for the corresponding actual label that were predicted as having a specific label. For each of the three tested GPT-4o configurations, a single normalized confusion matrix was obtained by averaging across that configuration's three above-mentioned runs.

2.4. Sentiment analysis

The best-performing approach, determined by overall accuracy and weighted F1 score, was employed to classify the remaining unlabeled tweets (n = 208,802) into one of our four sentiment categories. The human-labeled dataset, which was regarded as the most accurate reference (the "gold standard") in this study, was not relabeled by the best-performing approach. Instead, we combined the human-labeled dataset with the dataset labeled by the best-performing approach to create a completely labeled dataset, i.e., one in which all 231,777 tweets are associated with sentiment labels. All subsequent analyses were conducted using this combined dataset.

2.4.1. Differences in tweet length across sentiment categories

To explore the relationship between tweet length (measured by word count after removing URLs, usernames, and hashtags) and expressed sentiment, we conducted a one-way analysis of variance (ANOVA). Post-hoc pairwise comparisons were performed using Tukey's honestly significant difference (HSD) test.

2.4.2. Longitudinal change

After calculating monthly tweet frequencies for each sentiment, we followed a similar approach as Chaudhary et al. (2023). That is, we first identified activity peaks ("significant increases in the number of tweets published on a specific date within a span of time": p. 7) by visually checking a time-series visualization. We then confirmed the validity of these peaks by cross-referencing the corresponding time intervals with real-world events or news archives.

2.4.3. Sentiment changes during the key events

We divided the tweets surrounding each identified key event into two subsets: 1) a pre-event period, covering the 15 days before the event day, and 2) a post-event period, covering the event day and the 14 days following it. For each event, we then analyzed changes in the proportion of each sentiment category across these two periods. Following Lu et al. (2022), we conducted a Z-test to assess the significance of differences in sentiment proportions between the pre-event and post-event periods. A two-sided test was used to calculate p-values with the significance level set at 5%. To control for multiple comparisons, we applied a Bonferroni correction, i.e., adjusted the p-values according to the number of tests conducted. Since the Z-test requires a sample size >30 (Du et al., 2018), we performed statistical analysis only for sentiments where both the pre-event and post-event tweet counts exceeded 30. It is worth noting that this statistical test determines the significance of sentiment shifts, unlike the inter-coder reliability metrics reported earlier which assessed annotation consistency.

3. Results and discussion

This study's findings make an important contribution to methodological benchmarking by evaluating the performance of generative LLMs against more commonly used sentiment-analysis methods. The results demonstrate the best-performing method's capacity to effectively map longitudinal online food-related discourse. Specifically, our application of that framework confirms that it can successfully detect longitudinal patterns such as baseline sentiment stability and clear fluctuations around major industry milestones.

3.1. Model comparisons

The evaluation results are shown in Fig. 1. FT GPT-4o achieved the highest performance, with an overall accuracy of 0.824 (SD = 0.0008) and a weighted F1 score of 0.826 (SD = 0.0008). BERT performed comparably, with an overall accuracy of 0.783 and a weighted F1 score of 0.786. ZS GPT-4o (CB) attained an overall accuracy of 0.754 (SD = 0.0021) and a weighted F1 score of 0.764 (SD = 0.0019), and the parallel figures for ZS GPT-4o (no CB) were 0.728 (SD = 0.0011) and 0.737 (SD = 0.0010). VADER performed the worst, with an overall accuracy of 0.489 and a weighted F1 score of 0.517. For the reasons explained above, BERT (in evaluation mode) and VADER were tested only once each, so SD was not applicable to either of them in this case.

The Krippendorff's α values for ZS GPT-4o (no CB), ZS GPT-4o (CB), and FT GPT-4o, calculated across the three runs of each approach, were 0.971, 0.975, and 0.992, respectively. These values demonstrate the notably high internal consistency of their respective sets of three runs.

Below, due to the significant performance gap between VADER and the other approaches we tested, we discuss VADER's performance first, highlighting its limitations in comparison to more advanced approaches. Then, we comprehensively compare BERT and the three GPT-4o configurations, supported by their respective confusion matrices.

3.1.1. VADER

As noted above, VADER – unlike BERT and FT GPT-4o – does not require task-specific labeled data. Therefore, using it saves the considerable time and other costs associated with codebook development and data labeling. However, it performed the worst among the tested methods in both overall accuracy and weighted F1 score. As can be seen in the confusion matrix (Fig. 2A), its classification accuracy was 59% for positive, 48% for negative, and 45% for neutral tweets, indicating relatively low performance across all three of the categories it was able to deal with. Specifically, 34% of human-labeled negative tweets and 43% of human-labeled neutral tweets were misclassified as positive. A violin plot of VADER's sentiment-score distribution across human-labeled sentiment categories (Fig. 2B) further confirms and explains these high misclassification rates. That is, the plot demonstrates that

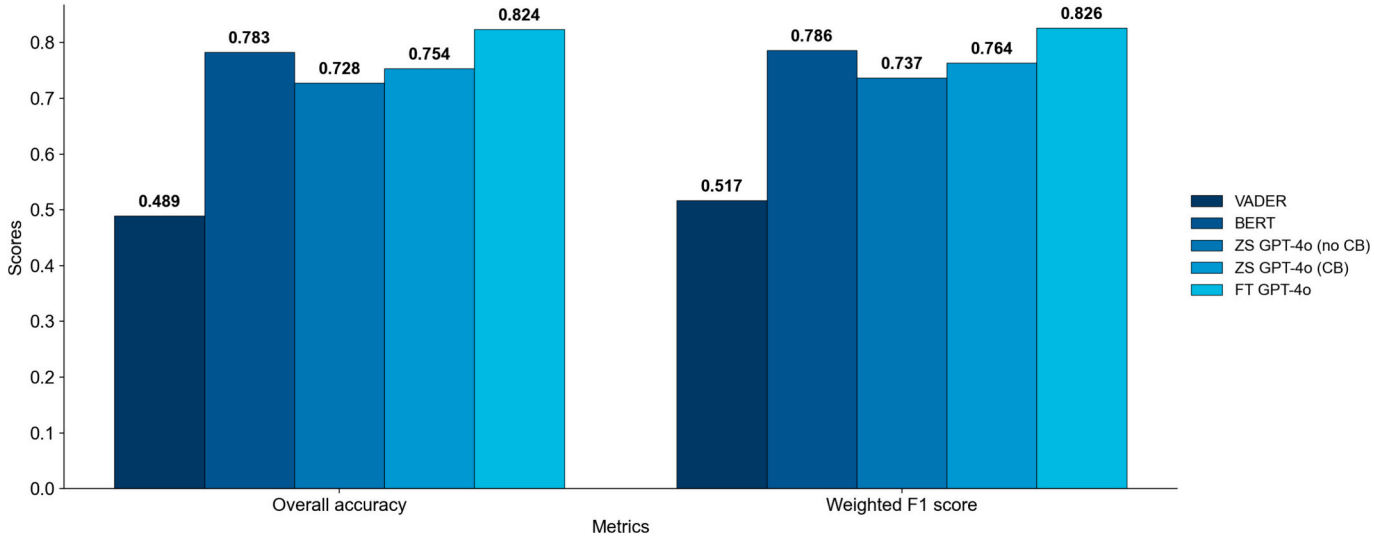


Fig. 1. Classification performance of the five tested sentiment-analysis methods. Error bars have been omitted because the variability was too small for them to be seen.

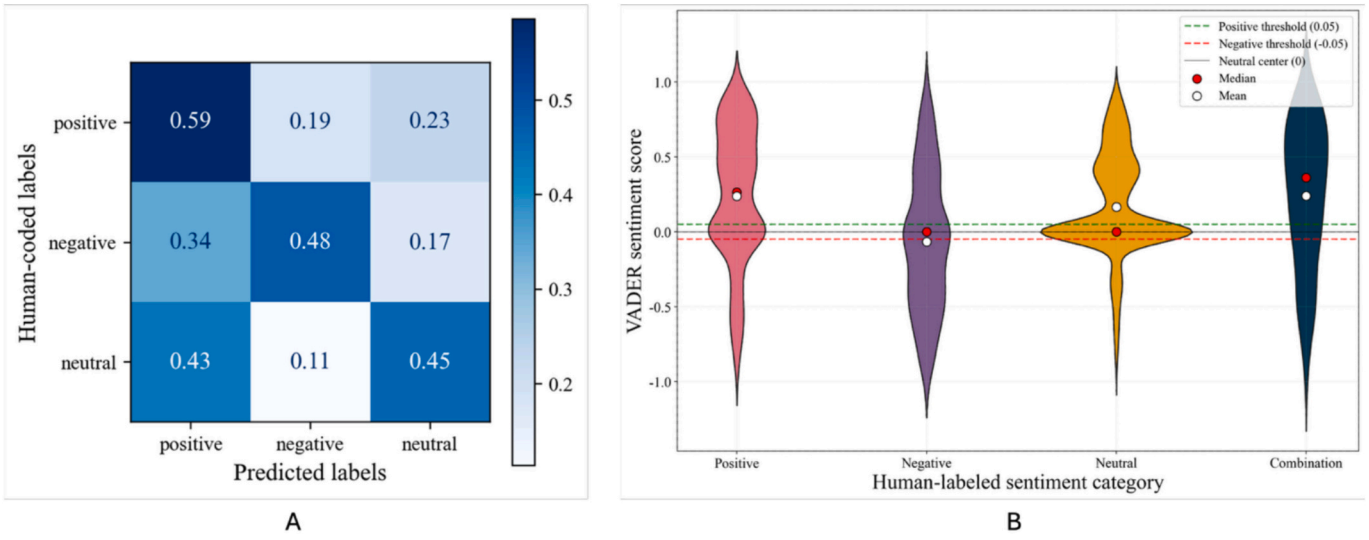


Fig. 2. Evaluation of VADER sentiment-analysis performance in the form of (A) a confusion matrix and (B) distribution of VADER's compound scores across human-labeled sentiment categories.

while human-labeled positive and human-labeled negative tweets showed a density peak within the neutral scoring band (±0.05), a noteworthy concentration of content that was human-labeled as neutral was scored by VADER as above the positive threshold.

This poor performance highlights some inherent limitations of lexicon-based methods. First, they rely entirely on pre-defined dictionaries, where each word or sentiment feature is assigned a fixed sentiment score (Mukhtar et al., 2018). However, in real-world language use, the meaning and sentiment of a word can change depending on its context. For example, the tweet "Lab-grown meat has major disruptive potential" was labeled as positive by annotators, but VADER classified it as negative, likely in part because it assigned a negative sentiment to the word "disruptive" based on its typical use in other contexts. Here, it should be noted that all example tweets presented in this paper are drawn from real data. Though they might have been lightly edited to enhance clarity and/or remove identifying information, their original meanings have not been modified.

Second, VADER calculates the overall sentiment of a sentence or longer document, and this renders it incapable of identifying the specific target of sentiment when multiple potential targets co-occur. For example, when cultured meat and other entities were mentioned in the same tweet, VADER struggled to ascertain which of them the expressed sentiment was directed toward. One such tweet was "Cultured meat will make conventional meat obsolete – no need to kill animals", which was labeled as positive by the human annotators but as negative by VADER. Another example tweet, "Interesting debate about the ethics of lab-grown meat happening. Check it out", was labeled by human annotators as neutral, but classified by VADER as positive. This is likely because VADER failed to recognize that the positive word "interesting" refers to the debate rather than to cultured meat itself. Similar incidences of failure to account for contextual differences might explain why, in our violin plot, a substantial concentration of VADER scores for human-labeled neutral tweets can be seen above the positive-sentiment threshold.

In summary, while VADER offers the advantage of quick deployment and low cost, its limitations in handling complex, multi-subject, or context-dependent sentiments made it the least reliable sentiment-analysis tool that we explored.

3.1.2. GPT-4o configurations and BERT

In this section, we first compare the performance of ZS GPT-4o (CB) against that of ZS GPT-4o (no CB) as a means of measuring the impact of highly detailed instructions (i.e., a codebook) on GPT-4o's performance. We then examine the performance of the two ZS GPT-4o configurations against FT GPT-4o and the well-established BERT model to ascertain how fine-tuning with labeled data affected model performance. Finally, we compare all three GPT-4o configurations against BERT and, based on the results, assess generative LLMs' potential as sentiment-analysis tools. These analyses focus on overall performance, as measured by overall accuracy and weighted F1 score, and the respective confusion matrices of the three GPT-4o configurations and BERT (Fig. 3).

Impact of a codebook. ZS GPT-4o (no CB) achieved classification accuracies of 81% and 76% for negative and neutral tweets, respectively (Fig. 3A). However, its performance on combination tweets was low (38%), with 40% of actual combination tweets misclassified as positive. ZS GPT-4o (CB), on the other hand, had a higher overall accuracy and a higher weighted F1 score (Fig. 1), as well as better accuracy across most categories, with "positive" being the exception (Fig. 3B). This variability could be related to prompt length, a double-edged sword: i.e., longer prompts tend to provide more information, but may also distract the model (Y. Chen et al., 2024).

Impact of fine-tuning. Our results show that both BERT and GPT-4o, when fine-tuned on our labeled data, outperformed both ZS GPT-4o configurations in overall accuracy and weighted F1 score. Roumeliotis et al. (2024) similarly found that, when tasked with sentiment analysis of cryptocurrency news, fine-tuned GPT-4 outperformed ZS GPT-4 that had received basic instructions. Our results suggest that, even when provided with detailed coding guidelines designed for human annotators, the performance of ZS GPT-4o will likely fall short of that achieved through fine-tuning with human-labeled data.

We also found that fine-tuning yielded the greatest accuracy improvements when sentiments were neutral or positive. Specifically, BERT achieved 73% accuracy for positive tweets and 84% for neutral ones (Fig. 3C), while FT GPT-4o achieved 78% and 86% accuracy for the same categories (Fig. 3D). Interestingly, BERT was less accurate when classifying negative and combination tweets than either of our ZS GPT-4o configurations. FT GPT-4o also performed worse than both ZS GPT-4o configurations in the negative category. In the combination category, it outperformed ZS GPT-4o (no CB) but fell short of ZS GPT-4o (CB). This underperformance of fine-tuning relative to ZS prompting in certain sentiment categories could have been due to the limited number of negative and combination tweets in our training data (respectively $n = 1717$, 9.3% and $n = 359$, 2.0%), given that fine-tuned models' performance heavily depends on the quantity of data used for fine-tuning (Wang et al., 2022).

Our observation that ZS GPT-4o underperformed BERT, irrespective of the presence or absence of a codebook, aligns with findings from other recent studies. For example, Bucher & Martini (2024) demonstrated that fine-tuned BERT-based models, such as RoBERTa, consistently outperformed ZS generative LLMs like GPT-3.5, GPT-4, and Claude Opus across various classification tasks (e.g., sentiment, party position) and text categories (e.g., news, speeches). Similarly, Lefort et al. (2024) found that, when fine-tuned, BERT-based models such as FinBERT and FinDRoBERTa could outperform GPT-3.5 and GPT-4 in ZS settings at sentiment analysis of financial news.

Our accuracy and weighted-F1 results imply that fine-tuning a generative LLM or a deep-learning model on a large amount of task-specific data could endow it with a superior ability to capture patterns and nuances in sentiment analysis, as compared to ZS LLMs. However, when training data in a specific category is limited – e.g., the combination category in our case – fine-tuning is likely to be less effective for it. In such circumstances, providing the generative LLM with a detailed and descriptive prompt, such as a codebook, might yield better

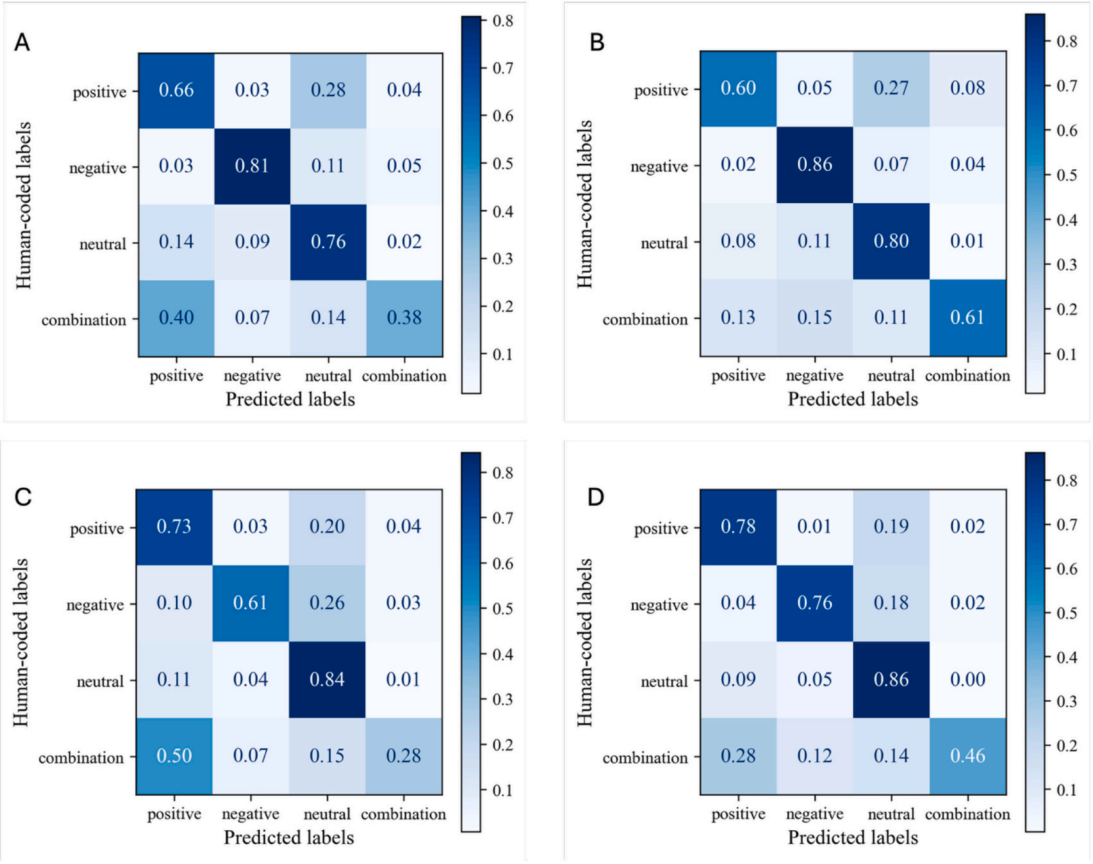


Fig. 3. Confusion matrices for (A) zero-shot GPT-4o without a codebook, (B) zero-shot GPT-4o with a codebook, (C) BERT, and (D) fine-tuned GPT-4o.

performance.

Methodological implications of generative LLMs as sentiment-analysis tools. Although fine-tuning yielded better overall performance than ZS prompting, its time and financial costs could be much higher. This discrepancy is primarily driven by fine-tuning's need for task-specific labeled data: a requirement that ZS approaches inherently avoid by relying solely on natural-language instructions. Crucially, the inference time for all the tested models was low when compared to the resource demands of either data labeling or the fine-tuning process. Specifically, both BERT and VADER were able to process our entire test set (i.e., 4595 tweets in BERT's case and 4492 in VADER's, for the reason mentioned above) in less than one minute when run on the free version of Google Colab on an NVIDIA Tesla T4 GPU with 16GB GPU RAM and 12GB system RAM. The inference time for the three GPT-4o configurations, when processing the same 4595 tweets, was consistently less than one hour. These sub-hour runtimes demonstrate that inference speed was not a limiting factor for any method. Rather, the more substantial resource demand might arise from the data-preparation phase, such as codebook development and manual annotation. Consequently, in practical applications, while raw runtime speed is of important consideration, it might be more critical to strike the right balance among annotation effort, cost, and accuracy, while still taking raw runtime speed into consideration.

Bearing this in mind, the performance of the tested ZS GPT-4o configurations was impressive: both outperformed VADER by a large margin, and performed comparably to BERT. Specifically, the overall accuracy of ZS GPT-4o (no CB) was approximately 5.5 percentage points lower than that of BERT, with a weighted F1 score 4.9 percentage points lower; and, when provided with a validated codebook, the overall accuracy of ZS GPT-4o was only 2.9 percentage points lower than BERT, and its weighted F1 score just 2.2 percentage points lower. Such performance gaps are acceptable, especially in light of BERT's robust NLP capabilities and the inherently subjective nature of sentiment analysis.

These results highlight the strong potential of ZS generative LLMs for sentiment analysis, particularly in the food domain, where labeled data tend to be difficult to access or produce. A key contribution of our work is that its pairing of human-developed and -validated, domain-specific definitions with an LLM to enable scalable sentiment labeling is generalizable. That is, future researchers should be able to use it to quickly uncover context-specific patterns in additional topical domains such as plant-based meat, among other food innovations, by incorporating their own domain-relevant constructs into it.

Importantly, however, adopting such an LLM does not eliminate the need for manual input. While we have confirmed that GPT-4o can achieve competitive performance without any task-specific labeled data, human input remains essential to 1) creating codebooks and revising them until agreement among annotators is achieved, and 2) generating the labeled data needed to evaluate model performance. It should also be borne in mind that the performance of ZS GPT-4o varied across sentiment categories. For example, both ZS configurations performed relatively well on negative and neutral sentiments, but less effectively on the positive and combination categories. If future studies require high accuracy across all sentiment classes, an additional round of manual review may be necessary for those categories in which generative LLMs underperform.

The FT GPT-4o we tested slightly outperformed BERT in overall accuracy and weighted F1 score, as well as in accuracy within each sentiment category. Notably, its accuracy was approximately 15 percentage points higher than BERT's for negative tweets and 18 percentage points higher for combination tweets, despite the limited quantity of training data for those two categories. This suggests that GPT-4o may generalize better from relatively small amounts of labeled data than BERT does. If so, such superiority is likely due to the former's much larger number of parameters and substantially greater quantity of pretraining data (Rehana et al., 2023): factors that may improve its ability to capture subtle distinctions and complex patterns in language.

Importantly, however, neither fine-tuning LLMs such as GPT-4o, nor performing inference with them, is cost-free; and the expense tends to increase with both the size of the training dataset and the amount of input and output data during inference (OpenAI, 2024d). When labeled data is limited, GPT-4o and other generative LLMs may be better equipped to learn task-specific patterns. When training data is ample, on the other hand, BERT and BERT-based models may be a more efficient choice due to their considerably lower computational-resource requirements and resulting financial costs. Researchers should therefore carefully weigh the expected performance gains from fine-tuning generative LLMs against the costs of both fine-tuning and inference.

Practical implications of broader NLP applications in the food industry. Further rapid advancements in generative LLM technology may decrease its costs while enhancing its capabilities. However, its full potential for the food industry remains underexplored. Ma et al. (2024) recently hailed LLMs' considerable potential for improving the efficiency and precision of processes throughout the global food system. We believe generative LLMs, either through fine-tuning or ZS prompting, hold strong promise not only for sentiment analysis and for various operational tasks, but also for a broad range of applications that could enhance the food industry's decision-making, efficiency, and innovation.

3.2. Sentiment analysis

3.2.1. Overview

As described in Section 2.4, the best-performing approach (i.e., FT GPT-4o) was used to classify the remaining unlabeled tweets which were combined with the human-labeled dataset to create the completely labeled dataset. Within this dataset, 146,933 tweets (63.4%) were classified as neutral, 57,477 (24.8%) as positive, 22,994 (9.9%) as negative, and just 4373 (1.9%) as combination. Crucially, these proportions primarily describe patterns in online discussion, and not necessarily the distribution of opinions in the broader population. That is, Twitter/X users are a self-selected and demographically skewed subset of the public (Statista, 2021), and active participation is disproportionately concentrated among a small fraction of those users (Hughes & Wojcik, 2019). This means that discussion of a given topic on the platform may reflect any combination of their interests/professional affiliations and the topic's visibility, as well as – or even instead of – the relative prevalence of opinions in society. Accordingly, the results should be interpreted as indicators of discourse composition, i.e., as reflecting the distribution of expressed attitudes within online discourse, not as direct indicators of population-level acceptance, rejection, or indifference. Their value lies in revealing how online discussion responds to real-world developments and how the social media framing of a novel food fluctuates over time.

Taking a closer look at the findings, the high proportion of neutral tweets does not necessarily indicate that most of the users whose tweets we sampled held neutral or balanced opinions about cultured meat. Rather, many of them simply did not include any sentiment toward cultured meat in their tweets, but instead shared information pertinent to such meat, typically as a brief, basic summary of an event or the headline of a news report (e.g., "Singapore Approves a Lab-Grown Meat Product, a Global First"). Nevertheless, the act of sharing such information demonstrates user engagement and participation in a broader conversation about cultured meat. This predominance of neutral sentiment may relate to the strong presence of media outlets on Twitter/X (Pindado & Barrena, 2021), one of the primary uses of which is news reporting (Java et al., 2007; Tandoc & Johnson, 2016). Moreover, during the time period examined by this study, cultured meat was only commercially available in Singapore (Räty et al., 2023), and it remained unavailable for commercial purchase in the United States. This limited availability likely contributed to its perceived novelty. The high proportion of neutral tweets we observed is also consistent with findings from previous social media-based studies on public attitudes toward

nuclear power (Kwon et al., 2024), vaccination (Tavoschi et al., 2020), and Bitcoin (Wahyuningsih & Chen, 2024), all of which reported that more than 50% of tweets were neutral. Those studies' definitions of "neutral" were similar to ours: i.e., they did not cover tweets that simultaneously contained both positive and negative sentiments.

When, following D'Andrea et al. (2019), we set aside the neutral tweets and only took sentiment-expressing ones into account, 67.7% of the remainder were positive, 27.1% negative, and 5.2% combination (Supplementary Materials, Fig. S1). This preponderance of positive tweets suggests a generally favorable sentiment toward cultured meat. However, relevant stakeholders should interpret these results with caution, and in particular, avoid using them to gauge consumer market potential. One important reason for such caution is that our results are based on numbers of tweets rather than numbers of individual users. In a previous project (T. Chen et al., 2024), we found that the key Twitter/X accounts driving the dissemination of information about this product category included many affiliated with cultured-meat companies (e.g., Upside Foods) and other organizations promoting it (e.g., the Good Food Institute, New Harvest). Given their promotional goals, it is highly likely that such accounts would disproportionately highlight its benefits, and this phenomenon could have contributed to a higher proportion of tweets being classified as positive than as negative.

Negativity has been shown to amplify news dissemination on social media, particularly through the re-sharing of user posts containing article links (Watson et al., 2024). Negative tweets, though less frequent in our data than positive ones, thus warrant attention due to their potential to more strongly impact public discourse: in particular, negatively influencing consumer attitudes by amplifying skepticism. Stakeholders aiming to promote novel foods should therefore actively participate in social media discussions and provide targeted responses to users' common concerns and questions.

3.2.2. Differences in tweet length across sentiment categories

One-way ANOVA revealed significant differences in tweet length across sentiment groups ($F(3, 231773) = 6853.2$, $p < 0.001$, $\eta^2 = 0.08$), with subsequent Tukey's HSD testing confirming that all pairwise differences were significant ($p < 0.001$). Specifically, neutral tweets were the shortest on average (M = 18.4 words, SD = 10.8), followed by negative (M = 24.8, SD = 13.4) and positive tweets (M = 25.2, SD = 12.9), while combination tweets were significantly longer than all other categories (M = 32.0, SD = 12.3).

These results may indicate that tweet length is associated with the structural complexity of the stance. Specifically, combination tweets could be longest because incorporating multiple viewpoints into a single post inherently requires more words than expressing a single-valence sentiment. The brevity of neutral tweets, meanwhile, might be consistent with their role in sharing inherently concise 'headline'-type information.

3.2.3. Longitudinal change

Longitudinal analysis enables identification of how a defined population's sentiment responds to specific events or shifts in discourse over time. Fig. 4 shows longitudinal changes in the tweet volumes associated with each sentiment. The minor peak in the quantity of positive tweets that can be observed in July 2016 coincided with the publication of a WIRED article titled "Lab-grown meat is coming to win over all you haters" (Cole, 2016), which highlighted this product's potential to address ethical and environmental issues, while also acknowledging challenges related to cost, taste, and public acceptance.

In February 2019, small peaks in both positive and negative tweets can be seen, coinciding with a tweet by American businessman William LeGate asking, "What are your thoughts on lab-grown meat?" This may highlight how public figures can stimulate both support and skepticism in online discussions, possibly by encouraging diverse perspectives or reigniting existing debates.

We also identified three much more prominent sentiment peaks corresponding to key real-world events: the cooking of the world's first cultured-beef burger at a conference in London on August 5, 2013 (Event 1) (BBC News, 2013); Singapore's approval of cultured meat for sale on December 2, 2020 (Event 2) (Reuters, 2020); and the completion of the U.S. Food and Drug Administration's first pre-market consultation for cultured meat on November 16, 2022 (Event 3) (FDA, 2022).

3.2.4. Sentiment changes during the events

As shown in Fig. 5, Event 1 was not associated with any significant change in the proportion of neutral tweets relative to the pre-event period ($p = 0.62$; all p-values in this paragraph are derived from Bonferroni-adjusted two-sided Z-tests). However, a 9.7 percentage point decline was detected in the proportion of positive tweets relative to the pre-Event 1 period ($p < 0.001$), accompanied by a rise of 7.4 percentage

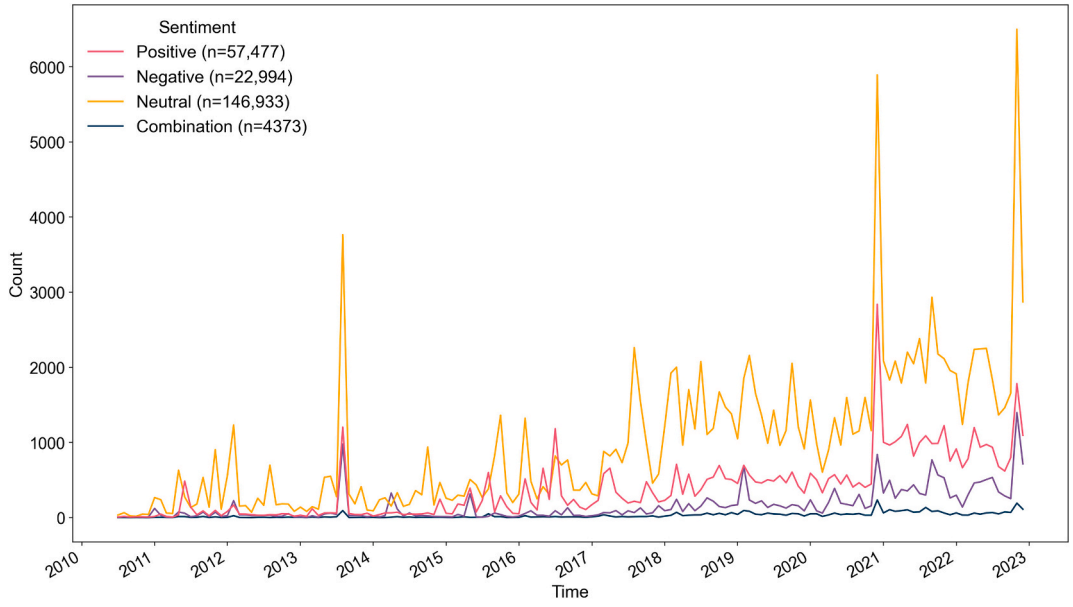


Fig. 4. Monthly volumes of each of the four target sentiments in tweets related to cultured meat. The three major and two minor peaks are annotated with corresponding events.

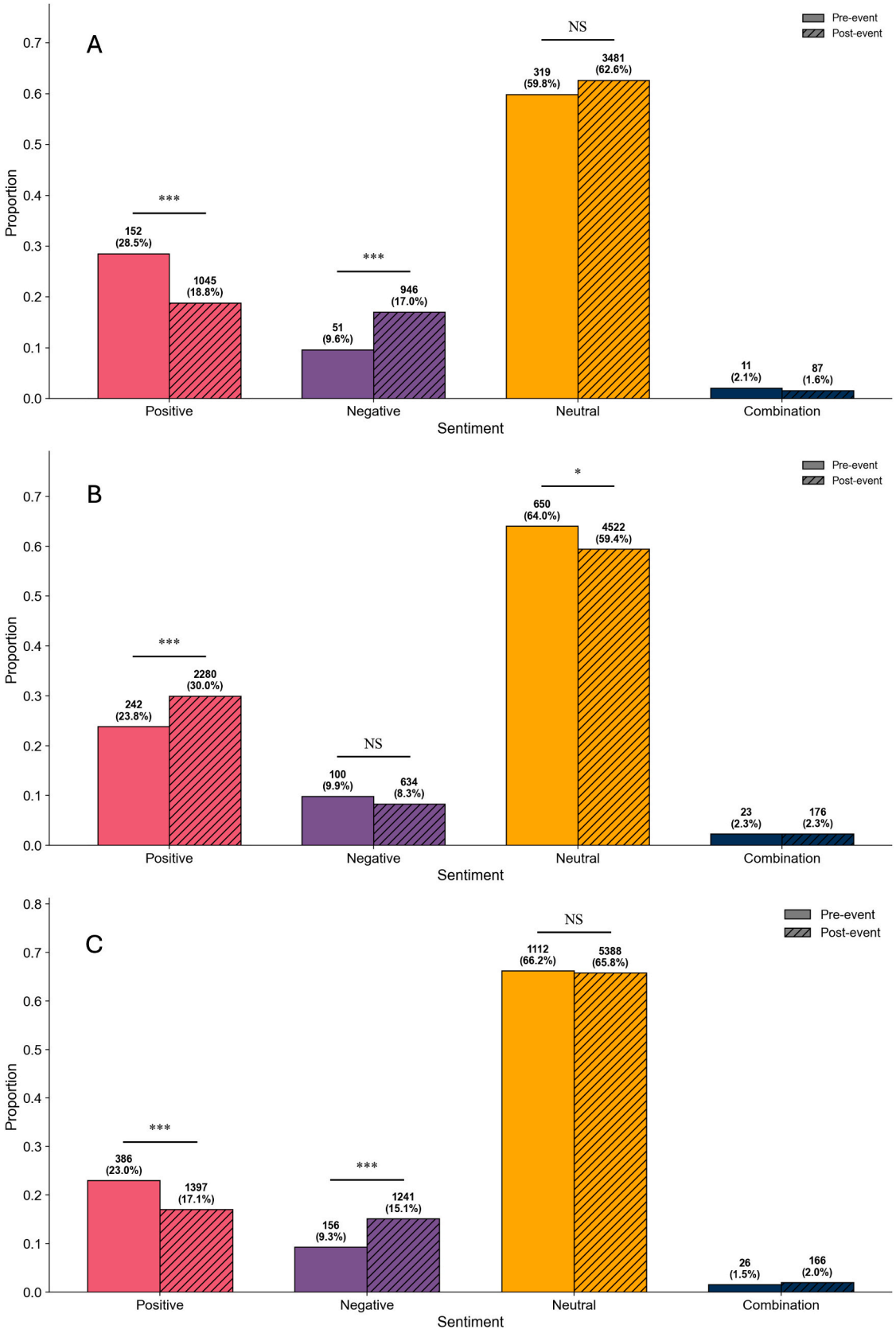


Fig. 5. Proportions of tweets expressing each of the four target sentiments during the pre- and post-event periods for (A) Event 1, (B) Event 2, and (C) Event 3. Statistical significance was established via two-sided two-proportion Z-tests with Bonferroni correction for multiple comparisons (NS = non-significant, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$). Z-tests were not conducted to examine the change during pre- and post-event periods for the combination sentiments category due to their limited sample size (see Section 2.4.3 for details).

points in the proportion of negative ones ($p < 0.001$). In the case of Event 2, the proportion of positive tweets rose significantly during the post-Event 2 period, by about 6.2 percentage points ($p < 0.001$), and the proportion of neutral tweets decreased, by 4.6 percentage points ($p = 0.014$). However, change in the proportion of negative tweets across the Event 2 window was non-significant ($p = 0.307$). Around Event 3, the proportions of neutral tweets also remained steady ($p = 1$), but positive tweets experienced a decline of 5.9 percentage points ($p < 0.001$), and negative ones, an increase of 5.8 percentage points ($p < 0.001$). Importantly, however, because Twitter/X data are not representative of the general population, these *Z*-tests should be interpreted only as indicators of whether the relative proportions of sentiment categories shifted enough to exceed random variation within this corpus, and not whether such shifts generalize to a broader public.

3.3. Limitations and future directions

While it has strongly demonstrated the utility of LLM-based sentiment analysis, this study has several limitations related to the computational methods employed and the nature of the data analyzed. First, GPT series models are closed-source (Chen et al., 2023), meaning that their weights and training data are not publicly available (Gao et al., 2023). This lack of transparency could introduce uncertainties. For example, updates and other changes to model architecture can cause the same prompt to produce performance fluctuations and inconsistent responses (Bhayana, 2024). Therefore, despite their strong performance in this study, future research referencing these generative LLMs' performance should ideally re-evaluate models' outputs before applying them to specific tasks. Because LLM APIs, including those in the GPT series, also often feature dynamically updated models (L. Chen et al., 2024), researchers should explicitly document the model version and parameter settings used. Such documentation will be necessary to ensure methodological clarity and allow future studies to properly contextualize or replicate any aspect of reported performance. It would be difficult to overstate the importance of methodological rigor, transparency, and continuous evaluation in LLM-based research, particularly given the frequent updates made by LLM vendors. Scholars must therefore be careful to select sentiment-analysis tools that are most appropriate for their data characteristics, disciplinary contexts, and research objectives.

Second, it is important to note that we used generative models simply as classifiers: i.e., constrained their output to one of four predefined sentiment categories, following a setup similar to those of commonly applied machine-learning approaches. As a result, their full generative capabilities were not leveraged. For example, beyond predicting sentiment labels, generative LLMs can produce confidence scores for their predictions (e.g., Groot & Valdenegro-Toro, 2024), which could facilitate more nuanced human evaluation of model responses. Future work could explore these richer outputs to enhance interpretability and human-in-the-loop validation.

Third, comparative cost analysis of the techniques we tested (and other similar ones) would be extremely valuable. However, it is hindered by a lack of transparent reporting. While surveys with large samples are well-known to be time-consuming and expensive (Zhang et al., 2022), it is not common practice to document and report the specific financial or time costs of nationally representative and other large-scale data-collection efforts. Similarly, we only recorded the inference time for each model and did not track either the precise time and financial costs associated with preliminary tasks such as codebook development and data labeling, or the direct API charges we incurred. We therefore encourage future researchers to comprehensively document and report both their runtime costs (e.g., tokens processed, API charges) and their human-resources costs (e.g., annotator time) so that cost-efficiency comparisons across social media analytics and survey-based approaches can be made transparently and cumulatively.

Fourth, social media data inherently present certain constraints. Specifically, tweets are often brief and informal (Fernández-Martínez, 2022), and our dataset confirms that many non-neutral tweets express support or opposition very simply (e.g., "I will never eat lab grown meat"), making it challenging to gain deeper insights into the underlying reasons for sentiments based solely on tweet content. Nevertheless, in situations where understanding a body of opinion is more important than uncovering the motivations behind it – e.g., decisions about product launches – Twitter/X and other social media platforms can serve as efficient and effective sources of timely guidance.

Finally, multiple selection mechanisms influence who participates in online conversation. Individuals who feel strongly about a topic are more likely than others to post about it, and participation rates on Twitter/X vary systematically across demographic and sociopolitical groups, a disparity that has been widening in recent years (Dinesh & Odabaş, 2023). Moreover, user subgroups that use formal terms like "cultured meat" and those using colloquial ones like "frankenmeat" may also differ from one another in other key respects (Bryant & Barnett, 2019). Our precision-oriented keyword strategy – optimized for relevance – may therefore have under-captured certain labels (e.g., "clean meat") that were preferred by certain user groups and/or at certain timepoints. While these biases limit the extent to which our sentiment proportions can be generalized to the broader public, they do not diminish our dataset's value for examining the tone of online discourse surrounding cultured meat and temporal trends in such discourse, including its participants' reactions to external events. Nevertheless, we recommend that future scholars broaden their keyword selection to encompass a wider spectrum of expressions and terminologies used by diverse or currently underrepresented user groups in online discussions of food topics.

4. Conclusions

In this methodological comparison of five sentiment-analysis approaches, FT GPT-4o achieved the highest overall accuracy and weighted F1 score, outperforming both BERT and VADER, while both ZS GPT-4o configurations (particularly the one with a codebook) achieved performance comparable to BERT at substantially lower annotation cost. These findings demonstrate that generative LLMs can provide scalable and accurate tools for analyzing large-scale, domain-specific social media data, particularly in contexts where pre-labeled datasets are limited and it might be infeasible for human coding alone to achieve adequate longitudinal or event-based coverage. Moreover, particularly in ZS configurations, generative LLMs offer ease of deployment: i.e., require no domain-specific fine-tuning and thus no labeled data, yet achieve accuracy comparable to fine-tuned deep-learning models such as BERT. This combination of ease and performance directly addresses a key methodological barrier in our field, where technical expertise and labeled data tends to be limited.

To demonstrate the practical utility of the best-performing LLM-based framework, i.e., FT GPT-4o, we applied it to a dataset of more than 200,000 cultured meat-related tweets. This efficiently yielded a descriptive overview of cultured-meat discourse: how much of it bears sentiment, its directional proportion when it does, and how specific milestones were associated with changes in those features. The specific patterns we discerned imply that approval events tend to amplify optimism, whereas announcements of regulatory scrutiny (in the absence of a declared outcome of such scrutiny) might trigger skepticism regarding safety and naturalness.

In sum, this study bolsters scholars' methodological toolkit for tracking shifts in online consumer narratives about emerging food technologies. As generative LLMs continue to evolve, their integration into consumer and market research offers promising avenues for near-real-time monitoring of online discourse surrounding novel foods and other food-related topics. However, continued methodological refinement – alongside attention to transparency, representativeness, and ethical data use – will be essential to ensuring that such models not only

perform accurately, but also yield insights that are meaningful for both the food-science and consumer-science communities, along with other stakeholders in the food system.

CRediT authorship contribution statement

Tianli Chen: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Leona Yi-Fan Su:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Yee Man Margaret Ng:** Writing – review & editing, Methodology, Funding acquisition. **Matthew J. Stasiewicz:** Writing – review & editing, Methodology. **Yi-Cheng Wang:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

The authors used generative AI during the preparation of this work. They reviewed and edited the content as needed, and take full responsibility for the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foodqual.2026.105968.

Data availability

The authors do not have permission to share data.

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